

In Program 2, the player controls a pufferfish by tapping the spacebar to swim and dodge the polluted ocean that is filled with plastic waste. The objective is to survive as long as possible to avoid colliding with the pillars of plastic. The score is shown while playing and the highest score is displayed in the menu. The interaction model is based on timing as the player has one input the spacebar to make the pufferfish increase buoyancy, while gravity pulls it downwards. This mechanic requires the player to carefully traverse the world using the motion to move through the narrow gaps. This simple but skill based game engages the player with the world and simulates the life of Puffy the pufferfish's life in a polluted world.

The two interactions I implemented are the swim mechanic which affects the buoyancy of Puffy and obstacle avoidance. When the spacebar isn't being pressed the pufferfish is pulled by gravity. This mechanic requires timing and control as pressing too early or late can cause a collision and trying to escape by attempting to go too far up or down will also cause the death of Puffy. I implemented a simple pillar design of plastic with narrow gaps for navigation and dodging. By passing through one pillar the score will increase and give a clear visual feedback loop where quick reaction and timing is rewarded with points and progression.

This interaction model can be reused throughout the game the further the player goes. Such as different types of pollutants or predators for future dodging mechanics. The swimming mechanics can be adapted for environmental changes such as deep ocean level would have less buoyancy per tap. I could have also added additional objectives such as collecting a token in between each obstacle for additional points and challenges for more skilled players. Overall this interaction is excellent as it combines easy and intuitive controls with the opportunity to become complex.